Tianhu Peng

PhD Researcher in Robotics

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Research Profile

PhD researcher at UCL specialising in **humanoid locomotion and expressive whole-body motion**. Combines robot learning, generative motion models, and model-based control, with experience in Unitree robot deployment and bipedal control at Tencent Robotics X.

Focus: humanoid locomotion; audio-conditioned motion generation; whole-body control; sim-to-real transfer.

Education

University College London (UCL) Nov 2024 – Present
PhD in Robotics London, UK

- Research on learning-based locomotion, expressive motion, and audio-conditioned dance on the Unitree G1 humanoid.

University of Leeds May 2023 – Oct 2024
Doctoral Research in Robotics; transferred to UCL Leeds, UK

- Completed first-year doctoral training and transfer viva; continued the research at UCL.

Imperial College London Oct 2020 – Oct 2021
MSc Human and Biological Robotics – Merit London, UK

University of Sheffield Sep 2017 – Jul 2020
BEng Mechatronics and Robotics Engineering – First Class Honours Sheffield, UK

Industrial Research Experience

Tencent Robotics X Laboratory Jul 2022 – Apr 2023
Robotics Researcher, Control Center Department Shenzhen, China

- Designed and implemented a model-based whole-body control framework for bipedal robots.
- Developed task-space tracking, contact constraints, force distribution, and hierarchical task prioritisation for coordinated whole-body motion.
- Integrated controllers with physical robot hardware and validated balance, walking, and motion execution.

Technical Expertise

Programming: Python, C++, PyTorch, ROS/ROS2, DDS, Git, Linux.
Robot learning: PPO, imitation learning, adversarial motion priors, curriculum learning, diffusion policies, teacher–student distillation.
Control: whole-body control, task-space tracking, contact dynamics, optimal control, sim-to-real deployment.
Platforms and tools: Unitree G1/A1, Isaac Gym, Isaac Lab, MuJoCo, Pinocchio, Unitree SDK.

Publications

Journal Articles

[J1] Z. Sun, L. Bao, **T. Peng**, J. Sun, and C. Zhou.
A High-Fidelity Digital Twin for Robotic Manipulation Based on 3D Gaussian Splatting.
Robot Learning, article 20260013, 2026.

[J2] L. Bao, J. Humphreys, **T. Peng**, and C. Zhou.
Deep Reinforcement Learning for Robotic Bipedal Locomotion: A Brief Survey.
Artificial Intelligence Review, vol. 59, article 38, 2026.

Conference and Workshop Papers

[C1] **T. Peng**, L. Bao, and C. Zhou.
Gait-Conditioned Reinforcement Learning with Multi-Phase Curriculum for Humanoid Locomotion.
IEEE-RAS 24th International Conference on Humanoid Robots (Humanoids), pp. 1087–1093, 2025.

[C2] M. Song, M. Zhang, A. Lyu, L. Bao, **T. Peng**, and C. Zhou.
OmniDexter: A Modular Tendon-Driven Robotic Wrist with Enhanced Precision and Versatility.
7th UK Robot Manipulation Workshop, 2025.

[C3] **T. Peng**, L. Bao, J. Humphreys, A. M. Delfaki, D. Kanoulas, and C. Zhou.
Learning Bipedal Walking on a Quadruped Robot via Adversarial Motion Priors.
Towards Autonomous Robotic Systems (TAROS 2024), LNCS 15052, pp. 118–129, 2024. **Best Paper Award**.

Submitted Manuscripts and Preprints

[P1] **T. Peng**, Y. Wang, L. Bao, Y. Jia, and C. Zhou.
AHEADance: Anticipatory Humanoid Dance from Streaming Audio.
Submitted to IEEE Transactions on Robot Learning (T-RL), 2026; arXiv:2611.13370. Project page.

[P2] L. Bao, H. Zhu, Y. Pan, J. Wang, J. Yu, **T. Peng**, Y. Jia, D. Kanoulas, and C. Zhou.
Beyond Kinematic Tracking: Preserving Expressive Characteristics in Humanoid Whole-Body Control.
arXiv preprint arXiv:2610.19393, 2026. Under review.

[P3] B. Long, L. Bao, **T. Peng**, J. Sun, and C. Zhou.
Reliability-Aware Sparse Route Memory for Round-Trip Vision-Language Navigation.
arXiv preprint arXiv:2609.34163, 2026.

[P4] Y. Pan, L. Bao, **T. Peng**, and C. Zhou.
PAMoR: Parameterized Affective Motion Generation in Real Time for Humanoid Robots.
arXiv preprint arXiv:2608.28213, 2026. Under review at IEEE Robotics and Automation Letters.

[P5] M. Ziegltrum, J. Jiao, **T. Peng**, C. Zhou, and D. Kanoulas.
Quadruped Parkour Learning: Sparsely Gated Mixture of Experts with Visual Input.
arXiv preprint arXiv:2604.19344, 2026.

[P6] M. Carroll, **T. Peng**, L. Bao, C. Zhou, and Z. Li.
SCDP: Learning Humanoid Locomotion from Partial Observations via Mixed-Observation Distillation.
arXiv preprint arXiv:2603.09574, 2026.

[P7] L. Bao, **T. Peng**, and C. Zhou.
Sim-to-Real Transfer in Deep Reinforcement Learning for Bipedal Locomotion.
Book chapter preprint, arXiv:2511.06465, 2025.

[P8] L. Bao, Y. Pan, **T. Peng**, D. Kanoulas, and C. Zhou.
Hierarchical Intention-Aware Expressive Motion Generation for Humanoid Robots.
arXiv preprint arXiv:2506.01563, 2025.

Full publication and citation record: Google Scholar.

Selected Research Projects

AHEADance: Anticipatory Humanoid Dance — Unitree G1

- Developed a streaming dance generator conditioned on heard audio and reliability-weighted forecasts of upcoming music (ListenAhead).
- Trained continuation from committed motion with Commit Forcing; deployed generated motion on a physical Unitree G1 through whole-body tracking.

Unified Humanoid Locomotion — Unitree G1

- Developed a recurrent policy for standing, walking, running, and gait transitions using gait-conditioned reward routing and a multi-phase curriculum.
- Demonstrated stable standing, walking, and transition behaviours on a physical Unitree G1 humanoid.

Bipedal Walking with Adversarial Motion Priors — Unitree A1

- Adapted adversarial motion priors and teacher–student learning to bipedal locomotion on a quadruped platform; evaluated flat and challenging terrains in simulation.
- Resulting first-author paper received the **TAROS 2024 Best Paper Award**.

Teaching Experience

University College London — 2025 – Present
Teaching Assistant – Legged Robot Systems — London, UK

- Support practical teaching and student debugging in locomotion control, simulation, reinforcement learning, and robot software integration.

Imperial College London — Oct 2021 – Dec 2021
Graduate Teaching Assistant – Robotics — London, UK

- Assisted students with robotics programming, technical problem solving, and laboratory exercises.

Awards, Talks, and Activities

- **Best Paper Award**, Towards Autonomous Robotic Systems (TAROS), 2024.
- Invited presentation at NVIDIA Robotics Day: robust humanoid locomotion via privileged distillation in Isaac Lab.
- ATEC Humanoid Robot Track: deployed a Unitree G1 for dynamic balance, outdoor traversal, and whole-body manipulation tasks.

Languages

Chinese – Native • English – Fluent

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